Local Tanh, Global Wall

Tensorization, Equality, and the Birkhoff–Dobrushin Divide

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Abstract

For a strictly positive finite Markov kernel P, Birkhoff's projective contraction coefficient is $\tanh(\Delta_H(P)/4)$, whereas Dobrushin's coefficient is the exact contraction coefficient for oscillation. The sharp comparison $\delta(P) \le \tanh(\Delta_H(P)/4)$ is known. We give a finite, explicit equality classification: apart from the rank-one case, equality holds exactly when a diameter-realizing pair of rows has a likelihood ratio taking two reciprocal values. On the closed probability simplex, the same classification holds inside a common face, while across distinct faces the infinite-distance bound is an equality exactly for disjoint supports. We then record the exact tensor law $\Delta_H(P \otimes Q) = \Delta_H(P) + \Delta_H(Q)$. Consequently the global Birkhoff coefficient of $P^{\otimes L}$ obeys hyperbolic addition and tends to one for every nonzero single-site diameter. The global Dobrushin coefficient also tends to one in every nontrivial product, while each coordinate oscillation contracts with the exact, volume-independent factor $\delta(P)$. For the binary Ising link all local quantities coincide at $\tanh|J|$, but the global projective coefficient is $\tanh(L|J|)$. Thus the same calculation both explains the ubiquitous local tanh and supplies a sharp wall against deducing a volume-uniform gap from the global tensor cone. We separate classical inputs, elementary tensor consequences, and the narrow equality and localization synthesis, and we include a reproducible exact-arithmetic kill test. A companion Lean module machine-checks the probability-normalized finite TV–Hilbert theorem with its equality classification and kernel lift, the singular disjoint-support equality branch, and the tensorization, hyperbolic-addition, and coordinate-averaging steps; the classical analytic Birkhoff–Hopf input remains external.

1 Introduction

Two contraction mechanisms are often placed side by side in finite-volume statistical mechanics. Birkhoff's theorem measures contraction in Hilbert's projective metric on a positive cone. Dobrushin's coefficient measures contraction of oscillations, or equivalently total variation between rows of a Markov kernel. Both produce a hyperbolic tangent in the binary Ising cell. It is tempting to read this coincidence as evidence that the global projective calculation supplies a volume-uniform mixing constant. The temptation is wrong in the most elementary independent product.

The purpose of this paper is to make the coincidence and the obstruction part of one theorem. There are three points.

(i) The sharp total-variation–Hilbert comparison is a pairwise statement. We record its complete equality condition on the closed simplex and lift it to finite kernels.

(ii) Hilbert diameter is exactly additive under tensor products. Hence the Birkhoff coefficient combines by hyperbolic addition, not by taking a maximum.

(iii) Dobrushin's global coefficient also degenerates on independent products. The volume-uniform statement lives instead in the vector of coordinate oscillations and, for interacting specifications, inside the Dobrushin interdependence window.

The wall is therefore not a caveat after the theorem. It is one of the theorem's conclusions.

Claim ledger

The classical projective contraction formula is due to Birkhoff [1] and was developed further by Bushell [2, 3]. The Markov coefficient and the interdependence method originate with Dobrushin [5, 6, 7]. Gaubert and Qu identify Dobrushin contraction with contraction in Hopf's oscillation seminorm on cones [8]. Reeb, Kastoryano and Wolf prove a general base-norm versus projective-diameter estimate [9]. Cohen and Fausti give the sharp total-variation–Hilbert inequality for probability measures, including boundary faces, noncomparable measures, and sharpness [4, Theorem 5.1 and Remark 15]. We do not claim any of these statements as new.

The narrow contribution of the present note is the following package: an if-and-only-if equality classification for finite distributions and kernels, including the closed-simplex singular branch; the exact tensor hyperbolic law placed next to that classification; and an exact global/local product dichotomy with the binary Ising saturation as a common test cell. The tensor identity itself is elementary. Cohen–Fausti state attainability, but we did not find the all-equality-cases statement in their paper. Our priority search found no source stating this entire finite-kernel package, but that search is not a claim of terminal priority.

2 Definitions and conventions

Let E be a nonempty finite set. For $u, v \in \mathbb{R}^E_{>0}$, Hilbert's projective distance is

$$d_H(u,v) = \log \frac{\max_{i\in E} u_i/v_i}{\min_{i\in E} u_i/v_i}. \tag{1}$$

It is a metric on rays, and it vanishes exactly when u and v are proportional.

For nonnegative probability vectors we use the standard extended convention: if $\operatorname{supp} p = \operatorname{supp} q$, formula (1) is evaluated on that common support; otherwise $d_H(p,q) = +\infty$. Thus $\tanh(d_H(p,q)/4) = 1$ for vectors in distinct faces.

For a strictly positive matrix $A = (a_{xy})_{x\in X, y\in Y}$, define

$$\Theta(A) = \max_{\substack{x,x'\in X\\ y,y'\in Y}} \frac{a_{xy}a_{x'y'}}{a_{xy'}a_{x'y}}, \tag{2}$$

$$\Delta_H(A) = \log \Theta(A). \tag{3}$$

Equivalently, $\Delta_H(A)$ is the largest Hilbert distance between two rows of A. Birkhoff's formula gives the projective contraction coefficient

$$\tau(A) = \tanh\left(\frac{\Delta_H(A)}{4}\right). \tag{4}$$

Transposition changes neither Θ nor Δ_H, so our row convention does not affect this number.

A Markov kernel P on E is row stochastic. Its Dobrushin coefficient is

$$\delta(P) = \frac{1}{2} \max_{x,x' \in E} \sum_{y \in E} |P(x,y) - P(x',y)|. \tag{5}$$

We use $\mathrm{TV}(p,q) = \frac{1}{2}\sum_y |p_y - q_y|$. Dobrushin's 1956 overlap convention is $1 - \delta(P)$, so formulas must be translated before comparing the two notations. For $f : E \to \mathbb{R}$, let

$$\operatorname{osc}(f) = \max_E f - \min_E f.$$

The standard dual formula says

$$\delta(P) = \sup_{\operatorname{osc}(f)>0} \frac{\operatorname{osc}(Pf)}{\operatorname{osc}(f)}. \tag{6}$$

3 The sharp pairwise comparison and all equality cases

The inequality in this section is known in a more general measure-theoretic form [4]. The finite proof is included because it exposes the equality condition needed later.

Theorem 3.1 (Sharp pairwise comparison)**.** *Let p, q be strictly positive probability vectors on a finite set. Put*

$$r_i = \frac{p_i}{q_i}, \qquad m = \min_i r_i, \qquad M = \max_i r_i.$$

Then

$$\mathrm{TV}(p,q) \le \frac{(M-1)(1-m)}{M-m} \le \frac{\sqrt{M}-\sqrt{m}}{\sqrt{M}+\sqrt{m}} = \tanh\left(\frac{d_H(p,q)}{4}\right), \tag{7}$$

with the middle expression interpreted as zero when $p = q$.

If $p \neq q$, equality between the first and last terms holds if and only if there is a number $t > 1$ such that

$$\left\{\frac{p_i}{q_i} : i \in E\right\} = \{t, t^{-1}\}. \tag{8}$$

In that case, with $A = \{i : p_i = tq_i\}$,

$$q(A) = \frac{1}{t+1}, \quad p(A) = \frac{t}{t+1}, \quad q(A^c) = \frac{t}{t+1}, \quad p(A^c) = \frac{1}{t+1}. \tag{9}$$

Thus the two measures exchange the masses of the two blocks.

Proof. Since $\sum_i q_i r_i = 1$, either $p = q$ or $m < 1 < M$. For every $r \in [m, M]$,

$$(r-1)_+ \le \frac{M-1}{M-m}(r-m).$$

Taking expectation under q gives the first inequality because $\mathrm{TV}(p,q) = \sum_i q_i(r_i - 1)_+$.

Write $a = \sqrt{M}$ and $b = \sqrt{m}$. With the positive common denominator $(a-b)(a+b)$, the numerator of the difference between the second and first bounds in (7) is

$$(a-b)^2 - (a^2-1)(1-b^2) = (ab-1)^2.$$

This proves the second inequality. The last identity follows from $d_H(p,q) = \log(M/m)$.

In the nontrivial case, equality in the chord bound requires every r_i to be an endpoint, so $r_i \in \{m, M\}$. Equality in the algebraic step requires $mM = 1$. Hence $M = t$ and $m = t^{-1}$ for $t > 1$. Conversely these two conditions make both inequalities equalities. Finally normalization gives $tq(A) + t^{-1}q(A^c) = 1$ together with $q(A) + q(A^c) = 1$, which yields (9). $\square$

Remark 3.2. *The equality condition is rigid in likelihood-ratio space but allows an arbitrary number of atoms in each of the two blocks. For two-point probabilities it simply says that the two vectors are nontrivial swaps.*

Corollary 3.3 (All equality cases on the closed simplex)**.** *Let p, q be nonnegative probability vectors on a finite set, and use the extended Hilbert distance above. Then*

$$\mathrm{TV}(p,q) \le \tanh\!\left(\frac{d_H(p,q)}{4}\right).$$

Equality holds exactly in one of the following mutually exclusive ways:

(a) $\operatorname{supp} p = \operatorname{supp} q$ *and either $p = q$, or the likelihood ratio on the common support takes two reciprocal values $\{t, t^{-1}\}$ for some $t > 1$;*

(b) $\operatorname{supp} p \cap \operatorname{supp} q = \varnothing$.

Proof. If the supports agree, restrict to their common face and apply Theorem 3.1. If they differ, the extended Hilbert distance is infinite and the right-hand side is one. For nonnegative normalized vectors,

$$\mathrm{TV}(p,q) = 1 - \sum_i \min(p_i, q_i).$$

Consequently $\mathrm{TV}(p,q) = 1$ exactly when every minimum vanishes, which is equivalent to disjoint supports. $\square$

4 Kernel comparison and equality classification

Theorem 4.1 (Birkhoff–Dobrushin comparison with equality)**.** *Let P be a strictly positive finite Markov kernel. Then*

$$\delta(P) \le \tau(P) = \tanh\!\left(\frac{\Delta_H(P)}{4}\right). \tag{10}$$

If $\Delta_H(P) = 0$, all rows of P agree and both sides vanish. If $\Delta_H(P) > 0$, equality holds in (10) *if and only if there are rows x, x' and a number $t > 1$ such that*

(a) $d_H(P(x,\cdot), P(x',\cdot)) = \Delta_H(P)$*; and*

(b) $P(x,y)/P(x',y) \in \{t, t^{-1}\}$ *for every y.*

Necessarily $t = e^{\Delta_H(P)/2}$.

Proof. Apply Theorem 3.1 to every pair of rows:

$$\mathrm{TV}(P(x,\cdot), P(x',\cdot)) \le \tanh\!\left(\frac{d_H(P(x,\cdot), P(x',\cdot))}{4}\right) \le \tau(P).$$

Taking the maximum proves (10). If equality holds, choose a row pair attaining $\delta(P)$. Both displayed inequalities must then be equalities. The second makes the pair diameter-realizing, and Theorem 3.1 gives the reciprocal two-valued likelihood ratio. The converse is immediate. For such a pair, its maximum-to-minimum ratio is t^2, so $2\log t = \Delta_H(P)$. □

Corollary 4.2 (Binary kernels)**.** *For a strictly positive two-state kernel with distinct rows, $\delta(P) = \tau(P)$ if and only if its rows are $(a, 1-a)$ and $(1-a, a)$ for some $a \neq \frac{1}{2}$.*

Proof. Theorem 4.1 says the two likelihood ratios are reciprocal. Equivalently $a(1-a) = b(1-b)$ for rows $(a, 1-a)$ and $(b, 1-b)$. Distinctness forces $b = 1-a$. □

The comparison can be strict even in dimension two. For example, the rows $(0.9, 0.1)$ and $(0.6, 0.4)$ have $\delta = 0.3$, while their likelihood ratios $3/2$ and $1/4$ are not reciprocal, so $\delta < \tau$.

Corollary 4.3 (Finite kernels with zeros)**.** *Let P be a finite nonnegative Markov kernel and let $\Delta_H(P)$ be the largest extended Hilbert distance between its rows. Then*

$$\delta(P) \le \tanh\left(\frac{\Delta_H(P)}{4}\right).$$

If all rows have a common support, equality is classified by Theorem 4.1 after restriction to that face. Otherwise $\Delta_H(P) = +\infty$, and equality holds exactly when two rows have disjoint supports.

Proof. Apply Corollary 3.3 pairwise and take maxima. In the infinite-diameter branch the right-hand side is one, while $\delta(P) = 1$ exactly when a maximizing row pair is disjoint. □

5 The tensor theorem and the projective wall

Theorem 5.1 (Exact tensor law)**.** *For strictly positive finite matrices A and B,*

$$\Theta(A \otimes B) = \Theta(A)\Theta(B), \qquad \Delta_H(A \otimes B) = \Delta_H(A) + \Delta_H(B). \tag{11}$$

Consequently

$$\tau(A \otimes B) = \frac{\tau(A) + \tau(B)}{1 + \tau(A)\tau(B)}. \tag{12}$$

In particular,

$$\Delta_H(A^{\otimes L}) = L\Delta_H(A), \qquad \tau(A^{\otimes L}) = \tanh(L \operatorname{arctanh} \tau(A)). \tag{13}$$

Proof. A cross ratio of $A \otimes B$ is the product of one cross ratio of A and one cross ratio of B. Hence it is at most $\Theta(A)\Theta(B)$. Because all index sets are finite, maximizing indices can be selected independently in the two factors, so the upper bound is attained. This proves (11). Equations (12) and (13) follow from Birkhoff's formula and the addition law for tanh. □

Corollary 5.2 (The global projective wall)**.** *If $\Delta_H(A) > 0$, then $\tau(A^{\otimes L}) \uparrow 1$. Therefore direct application of Birkhoff's theorem to the full positive tensor cone cannot produce a contraction factor $q < 1$ independent of L. If $\Delta_H(A) = 0$, then A has projective rank one and $\tau(A^{\otimes L}) = 0$.*

The statement is deliberately method-specific. It rules out a volume-uniform constant obtained from the global tensor-cone diameter. It does not rule out a different cone, a quotient adapted to locality, a weighted nonproduct seminorm, a block dynamics argument, or an independent spectral method. Each such change is a genuinely different mechanism.

Proposition 5.3 (Diagonal-weight blindness)**.** *If D_X, D_Y are positive diagonal matrices, then*

$$\Theta(D_X A D_Y) = \Theta(A), \qquad \Delta_H(D_X A D_Y) = \Delta_H(A).$$

Thus positive source and target weights cannot repair the tensor wall while the same projective cross ratios are used.

Proof. Every row factor from D_X and every column factor from D_Y occurs once in the numerator and once in the denominator of each cross ratio, and cancels. $\square$

6 Global Dobrushin degeneration and exact localization

The global wall is not unique to Hilbert's metric. Independent products separate distinct product measures, so their global total-variation coefficient also tends to one.

Theorem 6.1 (Global degeneration)**.** *Let P be a finite Markov kernel with two distinct positive rows p and q. Set their Hellinger affinity*

$$\rho(p, q) = \sum_y \sqrt{p_y q_y} < 1.$$

Then

$$\delta(P^{\otimes L}) \geq \mathrm{TV}(p^{\otimes L}, q^{\otimes L}) \geq 1 - \rho(p, q)^L. \tag{14}$$

Consequently $\delta(P^{\otimes L}) \to 1$.

Proof. The constant input words corresponding to the two rows select $p^{\otimes L}$ and $q^{\otimes L}$ as rows of the product kernel. For any probability vectors μ, ν,

$$1 - \mathrm{TV}(\mu, \nu) = \sum_z \min(\mu_z, \nu_z) \leq \sum_z \sqrt{\mu_z \nu_z}.$$

The affinity tensorizes, giving (14). Strict Cauchy–Schwarz gives $\rho(p, q) < 1$ when $p \neq q$. $\square$

The volume-uniform statement is coordinatewise. On E^L define

$$\mathrm{osc}_i(f) = \max_{x_{-i}} \left(\max_{a \in E} f(x_{-i}, a) - \min_{a \in E} f(x_{-i}, a) \right).$$

Theorem 6.2 (Exact local product contraction)**.** *For every $f : E^L \to \mathbb{R}$ and every coordinate i,*

$$\mathrm{osc}_i(P^{\otimes L} f) \leq \delta(P)\, \mathrm{osc}_i(f). \tag{15}$$

The constant is exact whenever $\delta(P) > 0$. Hence both $\max_i \mathrm{osc}_i$ and $\sum_i \mathrm{osc}_i$ contract by the same factor $\delta(P)$, independently of L.

Proof. Compare two inputs that differ only at coordinate i and condition on all output coordinates other than i. For each fixed outside output z_{-i}, the function $a \mapsto f(z_{-i}, a)$ has oscillation at most $\mathrm{osc}_i(f)$. The one-coordinate dual formula (6) bounds the resulting difference by $\delta(P)\,\mathrm{osc}_i(f)$. Averaging over z_{-i} and maximizing over the two inputs proves (15).

For exactness, choose a one-site function g attaining (6) and put $f(x) = g(x_i)$. Then both sides reduce to the one-site equality. The final two assertions follow by taking a maximum or a sum over i. $\square$

There is no contradiction between Theorems 6.1 and 6.2: the seminorms are different. The global oscillation can compare two words differing at every coordinate; osc_i permits one coordinate change at a time and retains the geometry needed by local comparison theory.

7 The binary Ising cell: exact coincidence and saturation

For $J \in \mathbb{R}$, consider

$$K_J = \begin{pmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{pmatrix}, \qquad P_J = \frac{1}{2\cosh J} K_J. \tag{16}$$

Proposition 7.1 (The common local tanh)**.** *For the binary link,*

$$\Delta_H(K_J) = 4|J|, \qquad \tau(K_J) = \delta(P_J) = \tanh|J|. \tag{17}$$

The nontrivial normalized eigenvalue of K_J is $\tanh J$*. Moreover, for the binary conditional*

$$\pi_h(+) = \frac{e^h}{e^h + e^{-h}} = \frac{1+\tanh h}{2},$$

one has, for $J \geq 0$,

$$\sup_{h \in \mathbb{R}} \mathrm{TV}(\pi_{h+J}, \pi_{h-J}) = \tanh J, \tag{18}$$

with equality at $h = 0$.

Proof. Diagonal entries against antidiagonal entries give the maximal cross ratio $e^{4|J|}$. The two rows of P_J are swaps, and their total variation is $|e^J - e^{-J}|/(e^J + e^{-J}) = \tanh|J|$. The eigenvectors $(1,1)$ and $(1,-1)$ give normalized eigenvalues 1 and $\tanh J$.

For the conditional calculation,

$$\mathrm{TV}(\pi_{h+J}, \pi_{h-J}) = \frac{1}{2}|\tanh(h+J) - \tanh(h-J)|.$$

The difference is even in h and decreases for $h > 0$; at $h = 0$ it equals $\tanh J$. □

Corollary 7.2 (Ising tensor saturation and wall)**.** *For $J \neq 0$,*

$$\tau(K_J^{\otimes L}) = \tanh(L|J|) \longrightarrow 1, \tag{19}$$

while the largest modulus below the top normalized eigenvalue of the product is $\tanh|J|$ *for every $L \geq 1$. The coordinatewise Dobrushin constant is also* $\tanh|J|$ *for every L.*

For completeness, put $a = (1 + \tanh J)/2$. Two antipodal rows of $P_J^{\otimes L}$ induce the laws $\mathrm{Bin}(L, a)$ and $\mathrm{Bin}(L, 1-a)$ after counting plus spins. Thus their total variation gives an exact finite sum lower bound for the global coefficient, and Theorem 6.1 yields the transparent estimate

$$\delta(P_J^{\otimes L}) \geq 1 - (\operatorname{sech} J)^L. \tag{20}$$

The spectral constant stays fixed while both global metric coefficients degenerate. This is the promised no-tension example: the obstruction belongs to the global instruments, not to slow mixing of the independent system.

8 From a link to the Dobrushin window

Let $\gamma_i(\cdot \mid x_{-i})$ be one-site conditional distributions and let $C = (c_{ij})$ be a Dobrushin interdependence matrix, so changing only site j changes the conditional at i in total variation by at most c_{ij}. The classical comparison theorem propagates local oscillations through C; see [6, 7] and the cone formulation in [8].

If

$$\alpha = \sup_i \sum_j c_{ij} < 1, \tag{21}$$

then the Neumann series $D = (I - C)^{-1} = \sum_{n\geq 0} C^n$ is uniformly bounded in the row-sum norm by $(1-\alpha)^{-1}$. If $c_{ij} = 0$ when graph distance $d(i,j) > 1$, then

$$D_{ij} \leq \sum_{n\geq d(i,j)} (C^n)_{ij} \leq \frac{\alpha^{d(i,j)}}{1-\alpha}. \tag{22}$$

This is where volume-uniformity lives: in a local vector inequality plus a subcritical row sum, not in the full tensor-cone diameter.

For a pairwise Ising specification, Proposition 7.1 gives the uniform envelope

$$c_{ij} \leq \tanh |J_{ij}|.$$

The supremum over the ambient real field is sharp, although the maximizing field need not be attainable in a particular finite boundary-value problem. On an anisotropic square lattice with horizontal coupling β and vertical coupling γ, the familiar sufficient window is

$$2 \tanh |\beta| + 2 \tanh |\gamma| < 1. \tag{23}$$

It is sufficient and generally not necessary. Nothing here extends a result beyond the Dobrushin window or proves optimality of that window.

9 What the wall rules out—and what it does not

Theorem 9.1 (Comparative theorem with its wall)**.** *For every strictly positive finite Markov kernel P:*

(1) Birkhoff's coefficient on the chosen global cone is exactly $\tau(P) = \tanh(\Delta_H(P)/4)$.

(2) Dobrushin's coefficient is exactly the oscillation contraction factor and satisfies $\delta(P) \leq \tau(P)$, with equality precisely as in Theorem 4.1.

(3) Under independent tensor powers, $\tau(P^{\otimes L})$ follows (13) *and tends to one unless P has identical rows. The global $\delta(P^{\otimes L})$ also tends to one unless all rows agree.*

(4) Each coordinate oscillation contracts with exact factor $\delta(P)$, uniformly in L. In interacting systems the analogous uniform comparison is controlled by (21)*.*

(5) For the binary Ising cell the three local constants—projective, Markov–Dobrushin, and conditional influence envelope—all equal $\tanh |J|$*, but the global projective tensor coefficient equals* $\tanh(L|J|)$*.*

Therefore no volume-uniform gap follows from applying the global Birkhoff diameter calculation to the tensor cone. This obstruction is part of the comparative conclusion.

The theorem rules out the proposed mechanism, not every projective method. A genuinely local cone, a quotient removing extensive directions, or a nonproduct Finsler structure might behave differently. Such a proposal must specify its cone and metric and pass the tensor test before a formalization campaign. Merely inserting positive diagonal Gibbs weights does not count, by Proposition 5.3.

10 Reproducibility and formalization status

The companion script `scripts/killtest_birkhoff_tensor_wall.py` uses exact rational arithmetic for

$$K = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

It verifies by enumeration through $L = 4$ that

$$\Theta(K^{\otimes L}) = 9^L, \qquad \tau(K^{\otimes L}) = \frac{3^L - 1}{3^L + 1},$$

whereas the normalized single-excitation eigenvalue remains $1/2$. A second exact-arithmetic certificate checks the pairwise envelope, equality and strict examples, cross-ratio tensorization, and coordinatewise saturation. It also checks 26,662 rational pairs in the closed three-point simplex, including all boundary faces, against the two-branch equality classification. These computations are regression tests, not substitutes for the proofs.

The companion module `YangMills/BirkhoffDobrushin/Wall.lean` formalizes the finite core that survived the kill test. In particular it checks: the scalar square identity and reciprocal-extrema equality condition; nonnegativity and endpoint rigidity of the chord slack, including its strictly positive finite weighted sum form; the equality of total variation with the positive-part likelihood expectation for normalized positive vectors; the sharp chord and square-root envelopes; the probability-normalized pairwise inequality with its complete if-and-only-if equality classification; its lift to a realized finite-kernel Dobrushin maximum; the overlap identity and the equivalence between total variation one and pointwise disjoint support for normalized nonnegative vectors; factorization of projective cross-ratios and preservation of a realized maximum under tensor products; the rational hyperbolic-addition law for Birkhoff coefficients; and the finite conditioning estimate that carries a fibrewise oscillation bound through averaging. The exact source was compiled with Lean `v4.29.0-rc6` on a Colab Pro+ CPU high-memory runtime. The successful package-import artifacts are:

Mathlib commit	`07642720480157414db592fa85b626dafb71355b`
source SHA-256	`0360a91a7061d9a47dbfec0dc2518175d05fb14edffacdf5b8452b1ebfa442cb`
module SHA-256	`6e37e9416c42dbdc9ab2667ea0843a5ba9bf34d7821c7a16c77cc583659cd919`

Both module compilation and a separate importing file had exit code zero. A second importing file explicitly checked the positive pairwise theorem, the closed-simplex singular branch, and the kernel comparison theorem, and materialized with SHA-256 `e0985e10eea3709f7907fb54fe1aa3532 f1298026c8d8524bb5324799c5cd8ce`.

The positive pairwise probability theorem, its realized-maximum kernel lift, and the disjoint-support equality branch are now formalized end to end. The reduction of a common boundary face to its positive support is retained as a finite mathematical reduction rather than a separate Lean theorem. This is still not an end-to-end formalization of every analytic statement in the paper: the general Birkhoff–Hopf contraction theorem, the logarithmic passage from maximal cross-ratio to Hilbert diameter, the Hellinger proof for the global product Dobrushin coefficient, and the interacting Dobrushin comparison theorem remain classical external inputs. Failed compiler and packaging passes are retained in the verification transcript rather than erased. No Lean or Lake process was run on the Windows owner desktop.

11 Limitations and conclusion

The analysis is finite. Zero entries are covered for the pairwise and kernel comparison through faces and the extended distance, but the tensor cross-ratio formalism itself is still stated on the strictly positive cone. The Dobrushin window is not claimed optimal. The local product theorem concerns independent coordinate updates; interacting dynamics require a specified update rule and comparison matrix. No Yang–Mills mass gap, infinite-volume spectral gap, or result outside the Dobrushin regime follows from this note.

Within those limits, the conclusion is sharp. The local binary cell explains the common tanh. Equality is completely visible in a two-valued reciprocal likelihood ratio. Tensorization then turns the same hyperbolic parameter into an extensive quantity and drives the global coefficient to one. Dobrushin's uniformity survives only after retaining coordinatewise information. The calculation and the wall are two sides of the same theorem.

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